

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T20 · Completing the Square

With Answers

Questions with full worked solutions

■ 19 pages

FAMILY MAP

Completing the Square

F01

Complete the square for a non-monic quadratic

8 Q | 24 marks

F02

Complete the square for a monic quadratic

1 Q | 3 marks

Working Unit 27

4MA1-2H Q25

ORIGINAL QUESTION

3 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

.....
(3)

Worked Solution 27

Complete the square

UNIT 1**3 marks****Rearrange**

$$28 + 24x - 6x^2 = 28 - 6(x^2 - 4x).$$

Complete

$$28 - 6((x - 2)^2 - 4) = 52 - 6(x - 2)^2.$$

Final answer

$$52 - 6(x - 2)^2 \quad (a = 52, b = 6, c = 2)$$

Question 16

4MA1-1H Q25

ORIGINAL QUESTION

4 marks

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a , b and c are constants.

$f(x) = \dots\dots\dots$

Worked Solution 16

Complete the square

MODULAR UNIT 1

4 marks

Rearrange

$$17 - 3x^2 + 12x = 17 - 3[(x - 2)^2 - 4].$$

Final answer

$$29 - 3(x - 2)^2$$

Question 24

4MA1-2H Q19

ORIGINAL QUESTION

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

Worked Solution 24

Complete the square

MODULAR UNIT 1

3 marks

Group the quadratic terms

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

Complete the square

$$3(x^2 - 2x) + 5 = 3[(x - 1)^2 - 1] + 5 = 3(x - 1)^2 + 2.$$

Final answer

$$3(x - 1)^2 + 2$$

Original question 15

4MA1-1H Q24 Q24

ORIGINAL QUESTION**4 marks**

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$

Worked solution

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maxi-

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 15 MODULAR UNIT 1
Chapter: Completing the Square 4 marks
Q24 – Chapter: Completing the Square
Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maxi-
mum form
Method
$q + 12x - qx^2 = -q[x^2 - (12/q)x] + q.$
Why: Factor $-q$ from the quadratic and linear x -terms.
Method
$x^2 - (12/q)x = (x - 6/q)^2 - 36/q^2.$
Why: Complete the square by halving the coefficient of x .
Method
$Therefore q + 12x - qx^2 = -q(x - 6/q)^2 + 36/q + q.$
Why: Substitute the completed-square identity and distribute $-q$.
Method
Matching $a - b(x - c)^2$ gives $a = q + 36/q$, $b = q$, and $c = 6/q$.
Why: Compare the constant, outside coefficient, and horizontal-shift terms.
Final answer
a: $a = q + 36/q$
Final answer
b: $b = q$
Final answer
c: $c = 6/q$

FINAL ANSWER

Original question 16

4MA1-1H Q20 Q20(a), Q20(b)

ORIGINAL QUESTION**4 marks**

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

.....
(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point *A* is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of *A*

(.....,)
(1)

Worked solution 16

Chapter: Completing the Square

MODULAR UNIT 1

4 marks

Q20(a) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

$$7+12x-3x^2=7-3(x^2-4x).$$

Why: Factor -3 from the quadratic terms.

Method

$$x^2-4x=(x-2)^2-4.$$

Why: Complete the square.

Method

$$\text{Therefore } 7+12x-3x^2=19-3(x-2)^2.$$

Why: Substitute the completed square and simplify.

Final answer

completed square: $19-3(x-2)^2$

Q20(b) – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

The maximum occurs when $(x-2)^2=0$, giving $A=(2,19)$.

Why: The negative coefficient means the squared term is subtracted from 19.

Final answer

maximum point: $(2,19)$

(c) Express $4x^2 - 8x + 7$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

(3)

Worked solution

November 2021 | Question 17 | 3 marks

Family: Complete the square for a non-monic quadratic | Variant: Positive non-unit leading coefficient

METHOD / WORKING

1. (c)

$$2. 4x^2 - 8x + 7 = 4(x^2 - 2x) + 7$$

$$3. =4\big((x-1)^2-1\big)+7$$

$$4. =4(x-1)^2+3$$

$$5. \text{Answer: } (6y + 5)(y - 1), f = \frac{8w-3}{w+2}, 4(x-1)^2 + 3$$

FINAL ANSWER

$$(6y + 5)(y - 1), f = \frac{8w-3}{w+2}, 4(x-1)^2 + 3$$

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

Worked solution

November 2025 | Question 21 | 3 marks

Family : Completes the square for a monic quadratic | Variant : Monic quadratic $x^2 + bx + c$

METHOD / WORKING

Worked solution
November 2025 Question 21 3 marks
Family : Completes the square for a monic quadratic Variant : Monic quadratic $x^2 + bx + c$
Method
$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$
$= 5((x-2)^2 - 4) + 23$
$= 5(x-2)^2 - 20 + 23$
$= 5(x-2)^2 + 3$
Answer: $5(x-2)^2 + 3$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

19 $ABCED$ is a five-sided shape.

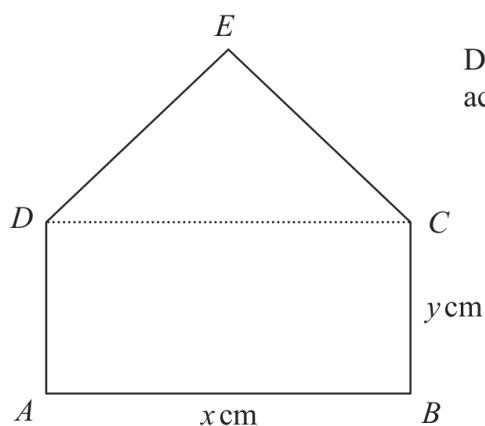


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} (200 - [6 - \sqrt{3}]x)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 2 marks

Family: Complete the square for a non-monic quadratic | Variant: Negative leading coefficient or maximum form

METHOD / WORKING

1. The perimeter is made from three sides of length x and two sides of length y :

2. $3x+2y=100$

3. So

4. $y = \frac{100-3x}{2}$

5. The area is the area of the rectangle plus the area of the equilateral triangle:

6. $R = xy + \frac{\sqrt{3}}{4}x^2$

7. Substitute $y = \frac{100-3x}{2}$:

8. $R = x(\frac{100-3x}{2}) + \frac{\sqrt{3}}{4}x^2$

9. $R = 50x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$

10. $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$

11. For the maximum, write

12. $R = -\frac{6-\sqrt{3}}{4}x^2 + 50x$

13. The maximum occurs at

14. $x = \frac{-b}{2a} = \frac{-50}{2(-\frac{6-\sqrt{3}}{4})}$

15. $x = \frac{100}{6-\sqrt{3}}$

16. The coefficient of x^2 is negative, so the quadratic graph opens downwards and this vertex gives a maximum.17. Answer: $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$, and $x = \frac{100}{6-\sqrt{3}}$.

FINAL ANSWER

$$R = \frac{x}{4}(200 - (6 - \sqrt{3})x), \text{ and } x = \frac{100}{6-\sqrt{3}}$$

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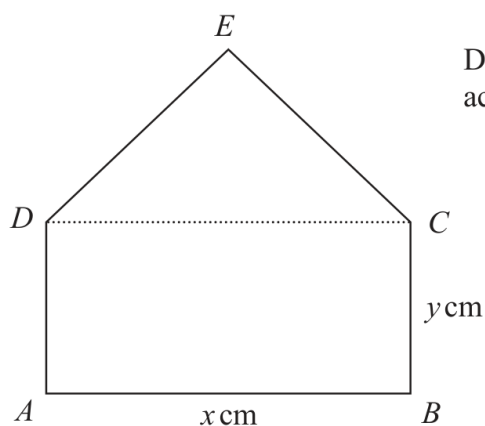


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(3)

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Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 1 marks

Family: Complete the square for a non-monic quadratic | Variant: Negative leading coefficient or maximum form

METHOD / WORKING

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3. So

4. $y = \frac{100-3x}{2}$

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FINAL ANSWER

$$R = \frac{x}{4}(200 - (6 - \sqrt{3})x), \text{ and } x = \frac{100}{6-\sqrt{3}}$$